Republic of the Philippines
CAVITE STATE UNIVERSITY
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COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY

ColTrace: Development of Mobile Application for File Tracking System of a College

Application Development and Emerging Technologies
Submitted to **John Aimiel Peña**
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Finals requirement
for the course Introduction to Computing
Bachelor of Science in Information Technology

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1. INTRODUCTION

1.1. Vision

The submission files may be completed manually or through paper-based documents such as letters, reports, records, and requests passed from various departments that lack proper tracking. It might require continual personal follow-up by staff and students, wasting valuable time and effort. One of the major issues facing the college is document delays, and there is no proper tracking of documents due to the lack of a file tracking system, which may result in documents being forgotten, processed slowly, or misplaced.

The development of a mobile application for a file tracking system is important because it allows users to track their documents, reduce file delays, and avoid file loss. It also increases efficiency, accountability, and transparency.

The mobile app is designed to track files by monitoring submission status labels such as submitted, pending, and late. The authorized user can monitor the flow of files.

Also, check to see which files have been submitted or are still pending. However, the mobile application for file tracking and monitoring can only be used within the college.

Objectives

File Tracking Operation **Reduce Delays** **Improve Accountability**

The primary purpose of ColTrace is to cater file tracking operations, reduce delays, and improve accountability in document management. The app gives centralised and easy access to all the files submitted by the different departments. Therefore, it is easier to manage and track all the files in an organisation, which creates more efficiencies, is more visible to everyone, and increases the productivity of those involved in these processes.

The app was developed based on feedback received from a client who wanted an efficient system for managing the tracking and retrieval of submitted files among multiple departments. ColTrace allows users to easily keep track of the flow of documents, view submission status, and facilitate proper management and documentation of significant documents.

The future improvements or feature we envision is to have a mobile application where all of the features of a good and effective file tracking system are present. Furthermore, I acknowledge that the features that we have right now are just part of the beginning of VSCC Company’s ColTrace App for becoming a world’s top used application in terms of file tracking system.

1.2. Project Scope

Can do:

File Tracking

Track the flow of files across departments. Monitor submission status (submitted, pending, late).

Folder Management

Track the flow of files across departments. Monitor submission status (submitted, pending, late).

Submission Monitoring

Check which files have been submitted or are pending. Track departmental compliance with deadlines.

Centralized System

Accessible system for multiple departments. Consolidates all departmental records in one platform.

View Images

Can view files that are in an image format.

Can’t do:

Document Viewing

The system can’t view the documents in PDF format. It only stores and tracks the files that are uploaded for submission and record-keeping purposes.

Sending Notification

The application does not support the sending of notifications to users, it cannot automatically notify them about deadlines, pending submissions, or updates within the system.

Edit of Users Information

The application can’t edit or remove the user's information. Once an admin created and registered the account in the system, the stored personal and account details cannot be edited or deleted.

Download Files

The system does not support the downloading of files uploaded by users; they can only be stored and tracked within the platform and cannot be saved to a device through the system.

1.3. Goal of the Project

The general objective of the project is to develop a mobile application that provides an accessible and efficient tracking system for the College of Engineering and Information Technology, which helps them manage and track their files by ensuring that they have an up-to-date status of the files.

1.4. Summary of the Team Roles

Casaverde, Mike

- *Logos and Documents*
- *UI/UX Design*

Casino, Lanz Michael

- *UI/UX Design*
- *Backend*

Sampaga, Charls

- *UI/UX Design*
- *Backend*

Viñas, Rikki Mae

- *Documents*
- *Backend*

2. PROJECT OVERVIEW

2.1. Project Concept

2.1.1. Mobile App Name

ColTrace (College File Trace System)

ColTrace or also known as “College Trace” described how efficient, systematic, and good on tracing file inside an organization or college. From the word itself, ‘College’ for the target users while ‘Trace’ is a function that describes how the system works.

2.1.2. App Category (example: Health Tracking, Gym App, Tutorial App)

- Document Tracking
- File Management
- Report Monitoring

2.1.3. Supported Platform (example: Android/ iOS/ Tablet)

The supported platform that our mobile application can be used in iOS, Android, or cross-platform environments for mobile application.

2.2. Project Brief (explain what the app does and what users can expect when using it)

The mobile application is solely for management of files and reports. Additionally, one of its purpose is to have an organized yet systematic easy to use for submitting with an accountability. The users can expect a user-friendly interface with features that is specialized in managing and tracking their submissions like create, read, update, and delete operations. These features will be shown in the user dashboard such a specified folder for each departments or what department they are in to upload the files and reports they need to submit.

2.3. Project Objectives.

2.3.1. General Objectives (Explain the main aim of the app)

The main goal of the project is to create a mobile application that offers a convenient and effective file-tracking system for the College of Engineering and Information Technology, enabling users to easily monitor, organize, and maintain the current status of their submitted and pending documents.

2.3.2. Specific Objectives

Specifically, this study aims to:

1. To develop a mobile-based file tracking system that allows the user to monitor the status of files.
2. Maintain the efficiently in documents for department chair, college admin, and instructor or professor.
3. To reduce the delays, errors, and document loss by creating digital file tracking system.
4. To provide the accessibility, transparency, and accountability of files.
5. To developed a digital tracking system for the College of Engineering and Information Technology.

2.4. Project Tools (List the technologies used:)

- Java Script
- React Native
- React Naavigation
- Expo
- node js
- express
- mongoDB

- Visual Studio Code

3. PROJECT METHODOLOGY

3.1. App Design Workflow (Explain how the group planned the application (wireframes, brainstorming, task planning, etc.))

The VSCC Company has its subject in university that they need to provide a capstone topic and as they gather information and data to decide what topic should they pursue, VSCC met a potential client where it said that he needs a file tracking system for their office or college. So, we conducted a sprint meeting and asked what and how would the system works. Eventually, we decided to use that as tentative topic and use it here in this subject for our prototype proposal. We divided the task into 4 phases which are Phase 1 – Brainstorming, Phase 2 – Basic System Structure, Phase 3 – Finalization of Features and System Architectures, and lastly Phase 4 – System Design and Development.

According to Andaya et al. (n.d.), Agile is effective for record management systems because it allows the development team to build prototypes, test features early, and improve the system based on client feedback. This approach reduces errors and ensures that the final system meets the actual requirements of the organization.

Agile also encourages regular communication between developers and users, which helps identify problems early and prevents major system issues during deployment.

3.2. Project Development (**Narrate** how does the team develop the project, include some sample code snippets: UI components, routes, schema, API calls)

The team develop the project by following the four phase set by the team in order to develop the project. Furthermore, will provide a snippets of codes.

UI Components:

```
const styles = StyleSheet.create({
  container: { flex: 1, backgroundColor: "#ffffff" },
  header: {
    flexDirection: "row",
    paddingTop: 50,
    paddingHorizontal: 12,
    backgroundColor: "#242323",
    elevation: 0,
    alignItems: "center",
    justifyContent: "space-between",
  },
  headerLeft: {
    flexDirection: "row",
    alignItems: "center",
  },
  headerLogo: {
    width: 60,
    height: 60,
    opacity: 1,
    marginRight: 6,
  },
  welcome: { fontSize: 12, color: "white" },
```

Routes:

```jsx
import React from "react";
import { createStackNavigator } from "@react-navigation/stack";
import LoginScreen from "../screens/LoginScreen";
import AdminDashboard from "../screens/AdminDashboard";
import SemiAdminDashboard from "../screens/SemiAdminDashboard";
import UserDashboard from "../screens/UserDashboard";
import FolderUpload from "../screens/FolderUpload";
import FolderDetailsScreen from "../screens/FolderDetailsScreen";
import SubmissionDetails from "../screens/SubmissionDetails";
import DepartmentFolders from "../screens/DepartmentFolders";
import UsersManagement from "../screens/UsersManagement";

const Stack = createStackNavigator();

const AppNavigator = () => {
  return (
    <Stack.Navigator initialRouteName="Login">
      <Stack.Screen
        name="Login"
        component={LoginScreen}
        options={{ headerShown: false }}
      />
      <Stack.Screen
        name="AdminDashboard"
        component={AdminDashboard}
        options={{ headerShown: false }}
      />
```

Schema:

```jsx
mongoose
  .connect("mongodb://127.0.0.1:27017/trackingDB")
  .then(() => console.log("Connected to MongoDB!"))
  .catch((err) => console.error("MongoDB connection error:", err));

const userSchema = new mongoose.Schema({
  username: { type: String, required: true, unique: true },
  password: { type: String, required: true },
  role: String,
  name: String,
  department: String,
});
const User = mongoose.model("User", userSchema);

// Submission Schema (defined once)
const submissionSchema = new mongoose.Schema({
  folderId: { type: mongoose.Schema.Types.ObjectId, required: true },
  username: { type: String, required: true },
  fileName: String,
  filePath: String, // Path to the uploaded file
  fileType: String, // MIME type of the file
  notes: String,
  submittedAt: { type: Date, default: Date.now },
});
```

API Calls:

```
app.post("/login", async (req, res) => {
  try {
    const { username, password } = req.body;

    if (!username || !password) {
      return res
        .status(400)
        .json({ success: false, message: "Username and password required" });
    }

    const user = await User.findOne({ username, password });

    if (user) {
      return res.json({ success: true, user });
    } else {
      return res
        .status(401)
        .json({ success: false, message: "Invalid credentials" });
    }
  } catch (error) {
    console.error("Login error:", error);
    return res.status(500).json({ success: false, message: "Server error" });
  }
});
```

4. PROJECT OUTPUT

This figure 1 and 2 is the representation that the session starts for a user or who access the system. It designed that way because to emphasize the logo of the application and for user interface. It could be improve by enhancing the design of the loading screen for its to be more appealing and not boring.



Figure 1. Loading Screen

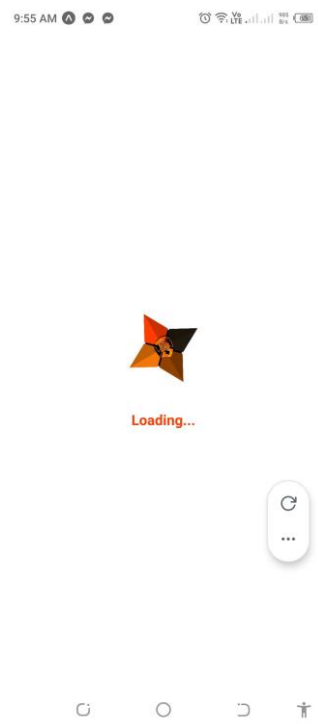


Figure 2. Loading Screen for Login Page

This figure indicates the login page of mobile application.

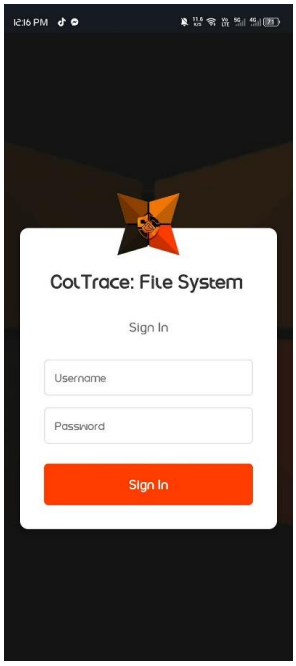


Figure 3. Login Page

This figure indicates the admin dashboard where the college office has the access to it.

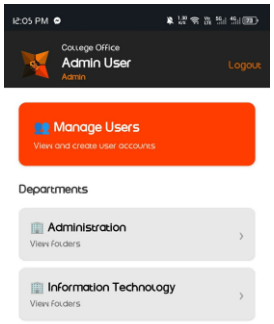


Figure 4. Admin Dashboard

This figure indicates the semi-admin of the system where the department chair has the access to it.

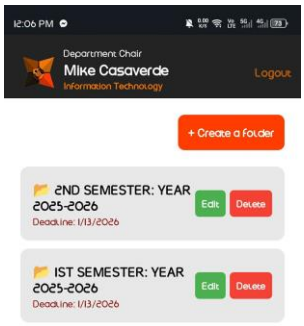


Figure 5. Semi-Admin Dashboard

This figure indicates the user dashboard where the instructors or any people inside the department has the access to upload and update files they uploaded.

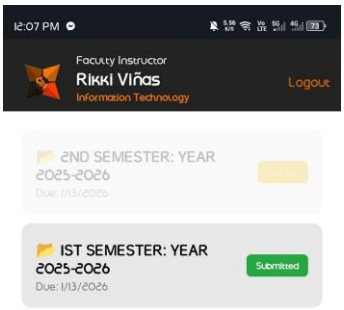


Figure 6. Users Dashboard

REFERENCES

Andaya, U.D. M., Baek, J., Cruz, M. R. S., & Ng, G. (n.d.). An agile methodology approach to a record management and monitoring system for Company B. University of Asia and the Pacific.

RUBRICS FOR DOCUMENTATION

Criteria	Excellent (4)	Good (3)	Fair (2)	Needs Improvement (1)	Score
Completeness of Sections	All required sections are complete and fully developed. No missing components.	Most sections are present but some details are underdeveloped.	Some sections are missing or lack depth.	Many required sections are missing or incomplete.	
Clarity of Content & Explanation	Ideas are clearly explained and easy to understand; shows strong understanding of the project.	Explanations are generally clear with minor confusion.	Explanations are unclear or lack detail; meaning is sometimes hard to follow.	Content is confusing, vague, or does not explain the project well.	
Methodology & Development Narrative	Development process is fully narrated; includes accurate steps and sample code; shows genuine progress.	Development is explained but may lack example detail.	Development explanation is minimal or vague; little evidence of process.	No real explanation of development; appears copied or incomplete.	
Screenshots & Output Description	Screenshots are clear, labeled correctly, and descriptions thoroughly explain function and purpose.	Screenshots are clear but descriptions are basic or incomplete.	Screenshots are present but unclear or lack explanation.	No screenshots, or screenshots do not relate to project.	
Formatting, Grammar & Organization	Document is well-organized, neat, properly formatted, and free of grammar/spelling errors.	Minor formatting or grammar issues; still readable and organized.	Noticeable errors or layout problems but still understandable.	Poor formatting, many errors, and difficult to read.	
Team Roles & Contributions	Roles are clearly defined and all members contributed equally.	Roles are defined but contribution balance is unclear.	Some members contributed more than others; unclear role assignment.	No clear role division; appears to be done by one person.	
References (APA 7th)	All sources are cited correctly in APA 7th format.	Minor formatting mistakes in citations.	Citations are incomplete or missing in several places.	No citations or improper citation style.	